

STAREHE GIRLS' 2025 PREDICTION



POSSIBLE KCSE EXAM



233/3

CHEMISTRY

PAPER 3 (PRACTICAL)

TIME: 2¼ HOURS

SCHOOL..... SIGN.....

CONFIDENTIAL

INSTRUCTION TO SCHOOLS

The information contained in this paper is to enable the head of the school and the teacher in charge of chemistry to make adequate preparations for this year's chemistry practical Joint examination. **NO ONE ELSE** should have access to this paper or acquire knowledge of its contents. Great care **must** be taken to ensure that the information herein does not reach the candidates either directly or indirectly. Chemistry teachers **SHOULD NOT** perform any of the experiments in the same room as the candidates or make the results of the experiments available to the candidates or give any other information related to the experiments.

In addition to the equipment, apparatus and chemicals in an ordinary Laboratory, each candidate require:

- | | |
|--|-----------------------|
| 1. About 150cm ³ of solution H. | 5. Pipette 25 ml |
| 2. About 150cm ³ solution J | 6. 1 pipette filler |
| 3. About 80cm ³ solution K | 7. 2 Conical flask |
| 4. Burette | 8. Means of labelling |

9. 250ml volumetric flask
10. White tile
11. 250ml plastic beaker
12. tissue paper
13. Thermometer (-10-110°C)
14. 50ml measuring cylinder
15. 500 ml Distilled water in a wash bottle
16. 2 boiling tubes
17. Filter funnel
18. Test tube holder
19. 5 clean dry test tubes
20. 100 ml plastic beaker
21. Metallic spatula
22. About 0.5g solid U in a stoppered container.
23. About 0.5g Solid P in a stoppered container.
24. 1 label
25. About 0.5g of sodium hydrogen carbonate.

Access solutions

- ❖ 2M sodium hydroxide supplied with a dropper.
- ❖ 2M ammonia solution supplied with a dropper.
- ❖ 0.1M Lead (II) nitrate supplied with a dropper.
- ❖ 0.1M barium nitrate solution.
- ❖ 0.2M Potassium iodide.
- ❖ Acidified potassium manganate (VII)
- ❖ Acidified potassium dichromate (VI)
- ❖ Phenolphthalein indicator supplied with a dropper.
- ❖ Source of heat.

NOTES

1. 2M sodium hydroxide is prepared by dissolving 80 g of sodium hydroxide pellets in about 200 cm³ of distilled water and topping up to 1 litre of solution
2. 2M ammonia solution is prepared by diluting 112 ml of concentrated ammonia solution in about 200 cm³ of distilled water and top up to 1 litre.
3. 0.1 M lead (II) nitrate is prepared by dissolving 33.0 g of solid lead (II) nitrate in about 400 ml of distilled water and top up to 1 litre of solution.
4. 0.1 M barium nitrate is prepared by dissolving 26 g of solid lead (II) nitrate in about 400 cm³ of distilled water and top up to 1 litre of solution.
5. 0.2M Potassium iodide is prepared by dissolving 33g in about 400cm³ of water and dilute to one litre of solution.
6. Solid U is maleic acid.
7. Solid P is Zinc chloride.
8. Solution H is prepared by dissolving 55 ml of concentrated Sulphuric (VI) acid in about 600cm³ of distilled water and dilute to 1 litre of solution.
9. Solution J is prepared by dissolving 80 g of sodium hydroxide in 400 cm³ of distilled water and top up to one litre of solution using distilled water.

- 10.** Solution K is prepared by dissolving 25g of citric acid $\text{C}_3\text{H}_5\text{O}(\text{COOH})_3$ RFM=192 in about 400cm³ and dilute to 1 litre of solution. (0.13M Citric acid)
11. Acidified potassium manganate (VII) is prepared by dissolving 3.12g of potassium manganate (VII) in 200cm³ of 2M sulphuric (VI) acid and making it to one litre.
12. Acidified potassium dichromate (VI) is prepared by dissolving 25g of potassium dichromate (VI) in 400cm³ of 2M sulphuric (VI) acid and making it to one litre.

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NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES

- ❖ *Write your name, index number, name of the school and the date in the spaces provided.*
- ❖ *You are required to spend 15 minutes of the 2¼hrs reading through the paper and make sure you have all the apparatus and chemicals needed for the practical.*
- ❖ *Answer all the questions in the spaces provided after each question*
- ❖ *Electronic calculators and mathematical tables may be used*

❖ All working must be clearly shown where necessary.

For Examiners Use Only

Question	Maximum score	Candidate's Score
1	23	
2	10	
3	07	
Total	40	

1) You are provided with the following

- An acid labeled solution H
- 2.0M sodium hydroxide solution labelled solution J
- Solution K containing 25 g per litre of an alkanoic acid

You are required to

- Prepare a dilute solution of sodium hydroxide solution J
- Determine the:
 - i) Molar mass of alkanoic acid, solution K.
 - ii) Reaction ratio between sodium hydroxide and acid H

Procedure 1:

Using a pipette and pipette filler, place 25 cm³ of solution J into a 250 ml volumetric flask. Add about 150 cm³ of distilled water and shake well. Add more distilled water to make up to the mark. Label this as solution L. **Retain the remaining solution J for use in procedure 2.**

Fill the burette with solution K. Using clean pipette and pipette filler, place 25 cm³ of solution L in to a 250 ml conical flask. Add 2 drops of phenolphthalein indicator and titrate with solution K. Record your results in the table I. Repeat the titration two more times and complete the table.

Table I (4 marks)

	I	II	III
Final Burette Reading (cm ³)			
Initial Burette Reading (cm ³)			
Volume Of Solution K (cm ³)			

- a) Calculate:
- i) The average volume of solution K used. (1 mark)
-
-
-
- ii) Concentration of solution L in moles per litre. (1mark)
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-
-
-
- iii) Concentration of the alkanoic acid solution K in moles per litre. (1 mole of the acid reacts with 3 moles of the base.) (1mark)

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iv) Molar mass of the alkanoic acid

(1mark)

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PROCEDURE 2

Fill a clean burette with solution H. Place 5 cm³ of solution H in 100 ml plastic beaker. Measure the initial temperature of solution H in the beaker and record it in the table II. Using a 50 ml measuring cylinder, measure 25 ml of solution J. add it to solution H in the beaker and immediately stir the mixture with the thermometer. Record the maximum temperature reached in table II. Repeat the experiment with other sets of volumes of solutions H and J and complete the table 2 below

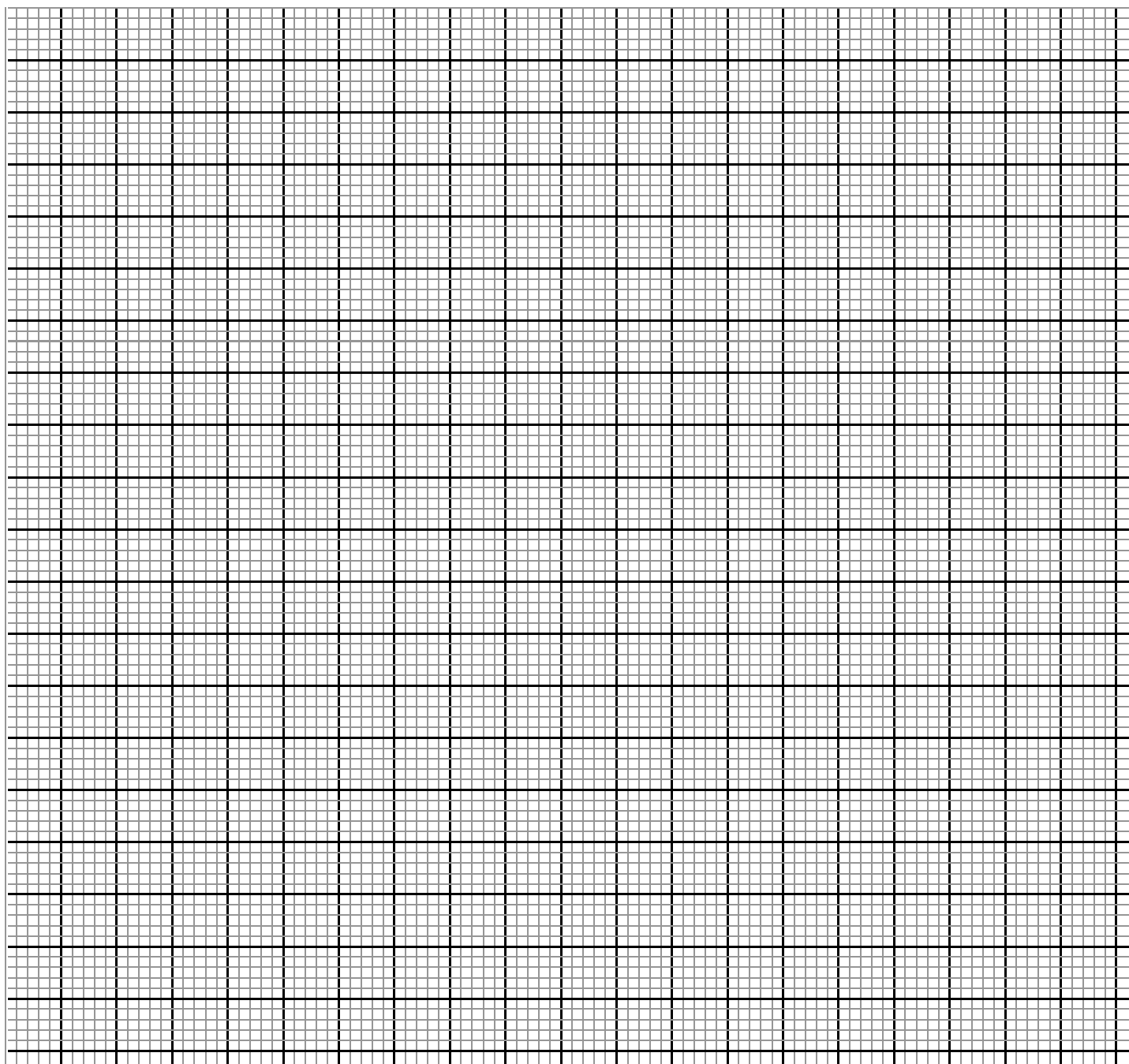
Table 2.

(6 marks)

Volume of solution H(cm ³)	5	9	13	17	21	25
Volume of solution J (cm ³)	25	21	17	13	9	5
Maximum temperature (°C)						
Initial temperature (°C)						

Change in temperature ($^{\circ}\text{C}$)						
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- a) On the grid provided, plot a graph of **change in temperature** against the **volume of solution H**.
(3 marks)



b) From the graph, determine the volume of solution H that was needed for complete neutralization.

(1 mark)

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c) Calculate the volume of solution J that fully reacted with the solution H stated in (b) above.

(1 mark)

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d) Calculate the:

i) Ratio between the volumes of solutions H and J that neutralized one another. **(1 mark)**

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ii) Concentration in moles per litre of the acid in solution H. (Assume that the volume ratio is the same as mole ratio). **(1 mark)**

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e) Calculate the molar heat of neutralization of sodium hydroxide solution in kJmol^{-1} given that the specific heat capacity for water is 4.2J/g/K and the density for water is 1g/cm^3 . **(2marks)**

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in excess.

Observation	inferences
1 mark	1 mark

d) To the third portion of the solution in a test tube, add 2M ammonia solution drop wise until in excess.

Observation	inferences
1 mark	$\frac{1}{2}$ mark

e) To the fourth portion of the solution in a test tube, add 2 - 3 drops of barium nitrate solution.

Observation	inferences
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$\frac{1}{2}$ mark	1 mark
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f) To fifth portion of the solution in a test tube, add 2 - 3 drops of lead (II) nitrate solution.

Observation	inferences
$\frac{1}{2}$ mark	$\frac{1}{2}$ mark

3) You are provided with substance U. You are required to

-Carry out tests on the substance U

-Record all observations and inferences accordingly.

½ mark	½ mark
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(iv) To the second portion of U in a test tube, add acidified potassium manganate (VII).

Observation	Inferences
1 mark	1 mark

(v) To the third portion of U in a test tube, add acidified potassium dichromate (VI) and warm the mixture.

observation	Inferences
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1 mark	½ mark
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